

MALLORY GIESIE

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EDUCATION

Master of Environmental Data Science, 3.9 GPA (June 2023)

Bren School of Environmental Science & Management – University of California, Santa Barbara (UCSB)

Highlighted Coursework: Machine Learning, Geospatial Analysis and Remote Sensing, Environmental Modeling, Databases & Data Management

Bachelor of Science in Quantitative Environmental Science, 3.8 GPA (December 2021)

The American University of Paris, Paris, France

Minor: Computer Science

Honors/Awards: Magna Cum Laude Honors, Global Citizen Award

Highlighted Coursework: Languages and Data Structures, Numerical Problem Solving in the Environment, Introduction to Artificial Intelligence

EXPERIENCE

AI Solution Engineer – ICF, Remote (10/25-present)

- Built and deployed a Python-based RAG application (FastAPI backend, React frontend) on Azure Container Apps, integrating Azure AI Search to answer program-specific technical assistance questions.
- Built and deployed an Azure-hosted AI planning application using Azure AI Foundry multi-agent orchestration and the OpenFEMA API to generate user-specific disaster action plans.
- Drives company-wide generative AI adoption by training consultants on AI-enabled workflows, AI-assisted coding tools, and client-facing AI use case development.

Climate Change and Sustainability Technology Specialist – ICF, Remote (10/23-10/25)

- Designed and deployed an agent workflow for resume ingestion and RFQ-ready DOCX generation; selected for company-sponsored productization, promoted to production, and on pace to save ~2,000 manual-editing hours annually.
- Deployed an ML model to generate DC fast-charging site suitability scores using parcel-level land use, VMT, and EV-adoption trajectory data.

[Climate Futures for Ecohydrological Modeling of Emerging Fire Regimes in Central Coast California](#)

Role: Data Manager | **Client**: UC Disaster Resilience Network | **Master's Capstone Project** (12/22– 6/23)

- Designed a method to construct customized climate scenarios using CMIP5 Climate Projections
- Developed an RShiny app for custom climate scenario creation, serving as an input generator for the RHESSys-fire ecohydrological model
- Synthesized Cal-Adapt models to summarize future trends for the Santa Barbara region

Research Assistant – The University of Konstanz's Limnological Institute, Konstanz, Germany (6/21–9/21)

- Organized data management practices and implemented protocols to ensure principles of reproducibility
- Completed a common garden experiment to test two species of Zooplankton's ability to coevolve
- Designed a simulation of a 24-hour grazing experiment to develop a plausible model for observed evolved variation in population grazing rates among two species of zooplankton

SKILLS

Languages: Python, R, JavaScript

AI/ML: LangChain, OpenAI Agents SDK, Azure OpenAI, Azure AI Foundry, CrewAI, Databricks ML

Scientific Modeling: Environmental Modeling, Geospatial Analysis, Uncertainty Analysis

Cloud Engineering & DevOps: Azure App Service, Azure Container Apps, Docker, Git/GitHub, CI/CD pipeline automation

Platforms: Azure, Databricks, AWS

Additional Experience: Biological Technician – McCormick Biological, Bakersfield, CA (4/22-7/22); Administrative Assistant – Wood Guttman & Bogart, Tustin, CA (4/18–8/19); Server – Hotel Seven4one, Laguna Beach, CA (8/16–8/19); Harvest Worker – Fallbrook Winery, Fallbrook, CA (8/18– 11/18)